

RIPHAH INTERNATIONAL UNIVERSITY GULBERG GREEN CAMPUS



The Closet Fashion Assistant (AI-Based Smart Outfit Recommendation System)

Submitted By:

Name	Sap Id	Semester
Simaab Malik	54910	SE-06
Nimra Tariq	54909	SE-06
Nighat Naseem	53740	SE-06
Afifa Amna	55466	SE-06
Anosha Awan	56200	SE-06
Areeza Afridi	55254	SE-06

Course
Artificial Intelligence

Submitted To
Ma'am Shumaila Qayum

Date
01/03/2026

Table of Contents

Introduction.....	3
Agent Type & Justification	3
Justification:	3
PEAS Description.....	3
Environment Properties	4
PEAS Framework for The Closet Fashion Assistant	4
Agent Function (high-level)	5
Summary	5

Agent Design and Environment Analysis

1. Introduction

This document presents the agent design and environment analysis for '**The Closet Fashion Assistant**' project as per the AI course requirements. It includes the chosen agent type with justification, complete PEAS description, sensors/actuators details, and environment properties.

2. Agent Type & Justification

Chosen Agent Type: Utility-based Agent (with learning elements)

Justification:

The system needs to optimize multiple conflicting criteria (weather match, occasion suitability, no repetition, clean clothes only). A simple reflex or goal-based agent is insufficient because it cannot evaluate "how good" an outfit is. Utility-based agent assigns a numerical utility score to each possible outfit combination using A* search + scoring function. Additionally, it learns from wear history to improve future recommendations (avoids repetition), making it partially learning-capable.

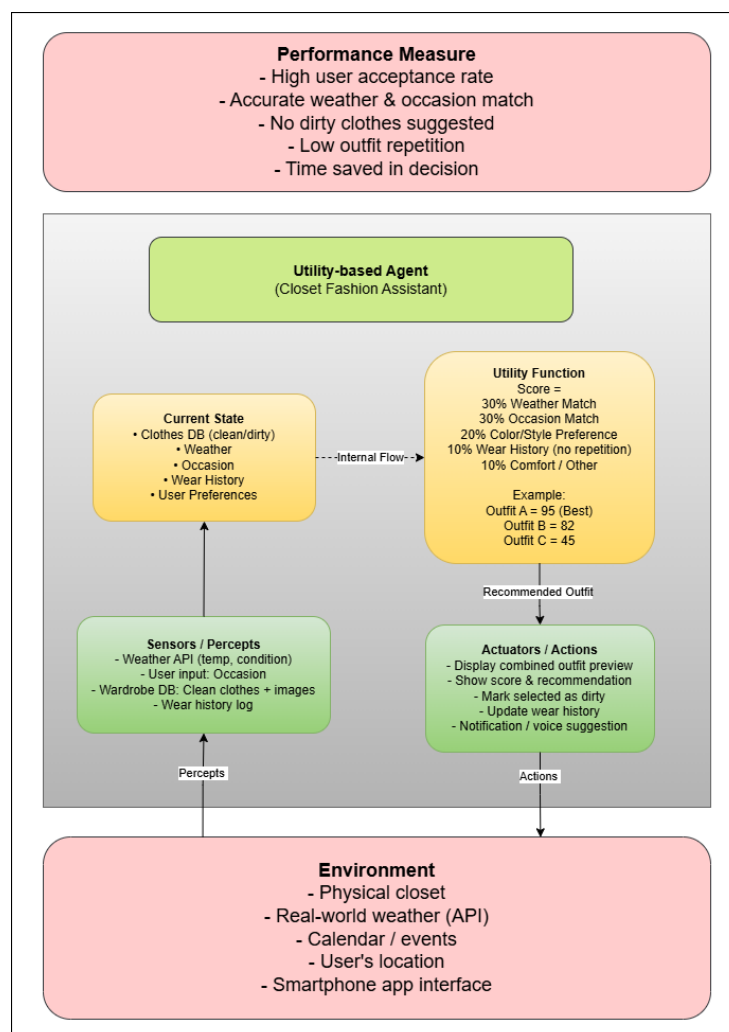
3. PEAS Description

Component	Description
Performance Measure	- High user acceptance rate of recommendations - Weather & occasion matching accuracy - Minimal repeated outfits - No dirty clothes suggested - Time saved in dressing decision
Environment	User's smartphone app + digital wardrobe database + real-world weather & occasions
Actuators	- Display combined outfit preview (shirt + pants + shoes image) - Mark selected clothes as "dirty" - Show alternative recommendation button - Laundry mode to mark items clean - Update wear history
Sensors	- Real-time weather data (via API: temperature, condition, season) - User input: occasion type (casual, formal, wedding, etc.) - Wardrobe state: uploaded clothes images + clean/dirty flags - Wear history log (previously worn outfits)

4. Environment Properties

Property	Category	Justification for Closet Assistant
Observable	Partially Observable	Agent only knows uploaded clothes & their status; real physical closet or forgotten items unknown
Deterministic	Stochastic	Weather can change suddenly; user may or may not accept suggestion (uncertain)
Episodic	Sequential	Each recommendation affects future state (dirty status changes, history updates)
Static	Dynamic	Real-time weather updates; clean/dirty status changes after use
Discrete	Discrete	Finite number of clothes & combinations in wardrobe
Single-agent	Single-agent	Personal stylist for one user only (no multi-user interaction)

PEAS Framework for The Closet Fashion Assistant:



Agent Function (high-level)

Percept → (weather, occasion, clean clothes list, history) → Generate possible outfit combinations → Use KNN for style similarity + A* with utility function (score = weather match + occasion match + cleanliness + diversity_from_history) → Select highest utility outfit → Display preview → Act based on user response

Summary:

The agent is designed as utility-based to handle multi-criteria optimization in a dynamic, partially observable environment. This design aligns with project goals of personalized, intelligent outfit recommendations.
